

2 The photograph shows the eyes of a meerkat.



(Source: Jeremy Pardoe / Alamy Stock Photo)

(a) Describe how the pupil of an eye dilates and contracts in response to changes in light intensity.

(4)

[illegible]

(b) Photoreceptors stimulate nerve transmission in the eye.

(i) Which of the following pairs of molecules forms the photoreceptor in an eye?

(1)

- ☐ **A** auxin and retinol
- ☐ **B** lysozyme and ribose
- ☐ **C** opsin and retinal
- ☐ **D** phytochrome and opsin

(ii) Which of these drugs would increase nerve transmission?

(1)

- ☐ **A** ecstasy (MDMA) and lidocaine
- ☐ **B** ecstasy (MDMA) and nicotine
- ☐ **C** lidocaine and nicotine
- ☐ **D** lidocaine only

(iii) In direct sunlight the diameter of the pupil was 4 mm.

In complete darkness the area of the pupil was 38.75 mm^2 .

Calculate the increase in diameter of the pupil when moving from direct sunlight to complete darkness.

(2)

Answer mm

(Total for Question 2 = 8 marks)